

MULTIMEDIA



UNIVERSITY

STUDENT ID NO

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MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER 1, 2017/2018

TCN2141 – COMPUTER NETWORKS

(All sections / Groups)

16 OCTOBER 2017
9.00 a.m. – 11.00 a.m.
(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This question paper consists of 5 pages with 6 questions only
2. Attempt **ALL SIX** questions.
3. All questions carry equal marks and the distribution of the marks for each question is given.
4. Please print all your answers in the answer booklet provided.

QUESTION 1

- a) Briefly describe what is NAT (Network Address Translation) and how it operates?
- b) Differentiate between multicasting and multiple unicasting.
- c) Given a host IP address with CIDR as 125.238.4.2/17. The administrator wants to create 512 subnets:
 - i. Find the subnet mask required.
 - ii. What is the current IP address subnet range?
 - iii. What is the subnet number for the 8th subnet?
 - iv. What is the subnet broadcast address for 15th subnet?
 - v. What are assignable addresses for the 25th subnet?
 - vi. What is 10th usable IP address in 35th subnet?

[2+ 2 + 6 =10 Marks]

QUESTION 2

- a) What are FOUR general techniques to improve quality of service (QoS) in a network?
- b) What is the difference between group-shared tree and a source-based tree in the context of multicast routing?
- c) Define the following terms used in Network Management: managing entity and managed device; management agent, MIB and network management protocol.

[4 + 2 + 4 =10 Marks]

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QUESTION 3

- a) Consider sending a 5500-byte datagram into a link that has an MTU datagram of 1500 bytes. Suppose the original datagram is stamped with the identification number 235411 as shown in table below (Original IP datagram). How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? Use IP fragments table below to give your answer.

Original IP Datagram					
Sequence	Identifier	Total Length	DF	MF	Offset
0	235411	5500	0	0	0

IP Fragments (Ethernet)						Range	
Sequence	Identifier	Total Length	DF	MF	Offset	Start	End

- b) Expand the following IPv6 addresses

- i. `::f53:6382:ab00:67db:bb27:7332`
- ii. `2819:af::35:cb2:b271`
- iii. `140:1abc:419a:a000::`

- c) Describe TWO categories of ICMP (Internet Control Message Protocol) messages.

[5+ 3 + 2 =10 Marks]

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QUESTION 4

- a) What is the purpose of an Address Resolution Protocol (ARP)?
- b) Compare Leaky Bucket and Token Bucket techniques. What is the purpose of both techniques?
- c) The following is a dump of TCP header in hexadecimal format.

05320017000000710000015A500207FF00000000

- i. What is the source port number?
- ii. What is the destination port number?
- iii. What the sequence number?
- iv. What is the acknowledgement number?
- v. What is the length of the header?

[2 +3 +5 = 10 Marks]

QUESTION 5

- a) Why do HTTP, FTP, SMTP and POP3 run on top of TCP rather than on UDP?
- b) When a router receives an IPv4 packet, it lowers the packet's time-to-live (TTL) value by one. If the TTL value drops to zero, the packet is discarded and an ICMP message is returned to the sender. How can this mechanism be used to trace the route a packet will follow when going from A to B?
- c) The difference between UDP and IP packets is minimal. Why shouldn't applications use IP directly?
- d) Running TCP across wireless networks may lead to performance problems. Why?
- e) Briefly describe the use of IGMP in multicast technology to save bandwidth?

[2+2+2+2+2 = 10 Marks]

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QUESTION 6

- a) Calculate IP header checksum if an IP header is given as below

4500002855704000800600000A0058143A1B5613

- b) Validated the above (a) checksum output at receiver's side and show that the IP header has no error.
- c) IPSec is an open standard framework for ensuring secure private communication over IP networks. Compare TWO protocols used to provide data confidentiality, authenticity, integrity, and replay protection.
- d) Distinguish between centralized and decentralized P2P networks.

[3+1+4+ 2= 10 Marks]

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